# EPSR-test-ex84 dockato: water docked into the 200 DMSO + 400 water box

Program(s): dockato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

## What this case tests

`dockato` [1] builds larger molecules from `.ato` building blocks: copies of the first molecule of a docking file are bonded onto free sites of a docked file, removing one atom of each to form the new bond. This case docks into a full simulation box, where the overlap test uses minimum-image distances.

## Code exercised

**Fortran (original, fortran_src/):**

- `dockato.f`: `dockato` (driver, parameters from command line or stdin), `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`
- `dockato.f`: `test_overlap`, `bond_molecule` (ROT/DIH groups), `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`
- `ato_manipulation_rev4.f`: `get_direct_bonds`, `get_rotgrp`, `centreofmass`, `ftype`; `ato_read_and_save.f`: `moloutc`; `minimagec.f`: `minimagec`

**Rust (port, rust/epsr-core/src/):**

- `tools/dockato.rs`: `run`, `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`, `test_overlap`
- `tools/dockato.rs`: `bond_molecule`, `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`, `get_direct_bonds`, `get_rotgrp`
- `ato.rs`: `State::moloutc` (ROT/DIH groups), `State::ftype`; `rng.rs`: `Ran1::ran1`

## Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 DMSO (atoms S, C, C, O, 6 H) and 400 water molecules (O1, H1, H1; SPC/E parameters [2]), 3200 atoms at 300 K in a cubic box of side 33.12 Å, a model of the DMSO-water mixtures of Soper and Luzar [3]. Atom `atomd` of the new molecule is placed on `atomv`, the molecule is rotated so that its `atomdr` neighbour lies on the `atomvr` neighbour of `atomv` (equal bond lengths required), an optional second (dihedral) rotation about the new bond uses `atomdrr`/`atomvrr` (random angle in [dminrr, dmaxrr] from `ran1`, the Park-Miller generator with Bays-Durham shuffle [4,5], or a fixed dihedral that creates a DIH group when dminrr = dmaxrr), overlaps are tested with a distance factor, `atomd` and `atomvr` are removed and the new rotational group, bonds and atom types are updated. A single water is bonded O-O onto one water of the box.

## Inputs

`WaterMolecule.ato` and `200DMSO400Water.ato` from EPSR-test, all parameters on the command line, one molecule per run (ndock = 1): run 1 dihedral 30-150°, factor 1.0; run 2 factor 0.6; run 3 overlaps allowed, fixed dihedral 60° (DIH group).

## How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Each run writes one new `.ato` file (coordinates, bonds, rotational groups, atom types and the random-number state), compared byte for byte with the stored Fortran output. With factor 1.0, hundreds of candidate water sites are rejected by `test_overlap` before one is accepted (README). Each output keeps 600 molecules and holds 3201 atoms (one water added, two H removed) and 401 rotational groups.

## Notes

Fortran behaviour reproduced: `nal(imv)` of the docked molecule is extended although the new atoms are at the end of the atom list, so molecule atom ranges overlap. With ndock > 1 on this box the Fortran writes `moltemplate(0)` (out of bounds); the port refuses that case, hence ndock = 1.

## References


[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).
[5] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.